

Question number	Answer	Additional guidance	Mark
8(a)	how much organic matter present in organisms (1)	ACCEPT amount / mass / content of / measure of dry mass / tissue containing carbon plant / animal / an organism / trophic level	(1)

Question number	Answer	Additional guidance	Mark
8(b)(i)	- total biomass of eukarya or all organisms calculated (1) 85.84 / 85.8 / 86 (%) (1)	468 / 545.2	(2)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	An answer that includes two of the following points: - cannot count all organisms as individuals (1) - organisms (within a group) are different {sizes / masses} (1)	ACCEPT take too long to count / too small to count IGNORE measure ACCEPT biomasses	(2)

Question number	Answer	Additional guidance	Mark
8(b)(iii)	An answer that includes three of the following points: Advantage: • very {visual / clear / easy} (way of presenting data / to understand) (1) • credit an example (1) Disadvantage: • cannot compare polygons with different shapes (but similar size) (1) • credit an example (1)	e.g easy to see the organisms with the highest biomass such as arthropods ACCEPT areas of polygons cannot be compared e.g. such as cnidarians and livestock	(3)

Question number	Answer	Additional guidance	Mark
8(c)	Indicative content: Voronoi diagram: • most biomass found on land and least found in marine (D) • because land provides the most suitable conditions to sustain living organisms* • such as light (for plants and algae) / water Plants: • all plants found on land (D) • because need enough light for photosynthesis • not in marine (D) • as salt water not suitable • not found deep underground (D) • as no light for photosynthesis Fungi: • predominantly found on land (D) • because they need a solid place to feed • because they need (gaseous) oxygen for respiration • not found deep underground (D) • because barely any other organisms live there to provide organic matter to decompose Protists: • found on both land and in marine environment (D) • because enough light for those that photosynthesise • they are the producers for the marine food chains* • can feed on {phytoplankton / algae} in the sea • not found deep underground (D) • because no light for those that photosynthesise • because no food Animals: • found on land and in a marine environment (D) • because have adapted for {feeding / movement} in both of these environment • can feed on {plants / seaweeds / other animals} • not found deep underground (D) • because no {food / water / pressure too high / ground too hard}	Level 1 : 1 mark = 2 descriptions 2 marks = 4 descriptions OR 1 {organism / Voronoi diagram} explained Level 2 : 3 marks = 2 {organisms / Voronoi diagram} explained 4 marks = 3 {organisms / Voronoi diagram} explained Level 3 : 5 marks = 4 {organisms / Voronoi diagram} explained 6 marks = 4 organisms explained AND explanation of Voronoi diagram* OR realises that there must be a non-plant producer in the sea*	(6)

	Bacteria: • one of two groups to be found in all three environments (D) • because can obtain energy from decomposition on the {land / marine} • can use alternative sources of energy deep underground Archaea: • one of two groups to be found in all three environments (D) • because can obtain energy from decomposition on the {land / marine} • can use alternative sources of energy deep underground	
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